

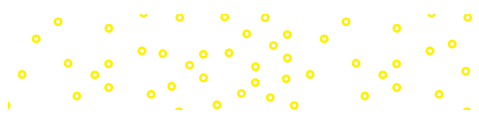
Conversion Factors

DIMENSION	METRIC	METRIC/ENGLISH
Acceleration	$1 \text{ m/s}^2 = 100 \text{ cm/s}^2$	$1 \text{ m/s}^2 = 3.2808 \text{ ft/s}^2$ $1 \text{ ft/s}^2 = 0.3048^* \text{ m/s}^2$
Area	$1 \text{ m}^2 = 10^4 \text{ cm}^2 = 10^6 \text{ mm}^2 = 10^{-6} \text{ km}^2$	$1 \text{ m}^2 = 1550 \text{ in}^2 = 10.764 \text{ ft}^2$ $1 \text{ ft}^2 = 144 \text{ in}^2 = 0.09290304^* \text{ m}^2$
Density	$1 \text{ g/cm}^3 = 1 \text{ kg/L} = 1000 \text{ kg/m}^3$	$1 \text{ g/cm}^3 = 62.428 \text{ lbm/ft}^3 = 0.036127 \text{ lbm/in}^3$ $1 \text{ lbm/in}^3 = 1728 \text{ lbm/ft}^3$ $1 \text{ kg/m}^3 = 0.062428 \text{ lbm/ft}^3$
Energy, heat, work, internal energy, enthalpy	$1 \text{ kJ} = 1000 \text{ J} = 1000 \text{ N}\cdot\text{m} = 1 \text{ kPa}\cdot\text{m}^3$ $1 \text{ kJ/kg} = 1000 \text{ m}^2/\text{s}^2$ $1 \text{ kWh} = 3600 \text{ kJ}$ $1 \text{ cal}^\dagger = 4.184 \text{ J}$ $1 \text{ IT cal}^\dagger = 4.1868 \text{ J}$ $1 \text{ Cal}^\ddagger = 4.1868 \text{ kJ}$	$1 \text{ kJ} = 0.94782 \text{ Btu}$ $1 \text{ Btu} = 1.055056 \text{ kJ}$ $= 5.40395 \text{ psia}\cdot\text{ft}^3 = 778.169 \text{ lbf}\cdot\text{ft}$ $1 \text{ Btu/lbm} = 25,037 \text{ ft}^2/\text{s}^2 = 2.326^* \text{ kJ/kg}$ $1 \text{ kJ/kg} = 0.430 \text{ Btu/lbm}$ $1 \text{ kWh} = 3412.14 \text{ Btu}$ $1 \text{ therm} = 10^5 \text{ Btu} = 1.055 \times 10^5 \text{ kJ}$ (natural gas)
Force	$1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2 = 10^5 \text{ dyne}$ $1 \text{ kgf} = 9.80665 \text{ N}$	$1 \text{ N} = 0.22481 \text{ lbf}$ $1 \text{ lbf} = 32.174 \text{ lbm}\cdot\text{ft/s}^2 = 4.44822 \text{ N}$
Heat flux	$1 \text{ W/cm}^2 = 10^4 \text{ W/m}^2$	$1 \text{ W/m}^2 = 0.3171 \text{ Btu/h}\cdot\text{ft}^2$
Heat transfer coefficient	$1 \text{ W/m}^2\cdot^\circ\text{C} = 1 \text{ W/m}^2\cdot\text{K}$	$1 \text{ W/m}^2\cdot^\circ\text{C} = 0.17612 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$
Length	$1 \text{ m} = 100 \text{ cm} = 1000 \text{ mm} = 10^6 \mu\text{m}$ $1 \text{ km} = 1000 \text{ m}$	$1 \text{ m} = 39.370 \text{ in} = 3.2808 \text{ ft} = 1.0926 \text{ yd}$ $1 \text{ ft} = 12 \text{ in} = 0.3048^* \text{ m}$ $1 \text{ mile} = 5280 \text{ ft} = 1.6093 \text{ km}$ $1 \text{ in} = 2.54^* \text{ cm}$
Mass	$1 \text{ kg} = 1000 \text{ g}$ $1 \text{ metric ton} = 1000 \text{ kg}$	$1 \text{ kg} = 2.2046226 \text{ lbm}$ $1 \text{ lbm} = 0.45359237^* \text{ kg}$ $1 \text{ ounce} = 28.3495 \text{ g}$ $1 \text{ slug} = 32.174 \text{ lbm} = 14.5939 \text{ kg}$ $1 \text{ short ton} = 2000 \text{ lbm} = 907.1847 \text{ kg}$
Power, heat transfer rate	$1 \text{ W} = 1 \text{ J/s}$ $1 \text{ kW} = 1000 \text{ W} = 1.341 \text{ hp}$ $1 \text{ hp}^\ddagger = 745.7 \text{ W}$	$1 \text{ kW} = 3412.14 \text{ Btu/h}$ $= 737.56 \text{ lbf}\cdot\text{ft/s}$ $1 \text{ hp} = 550 \text{ lbf}\cdot\text{ft/s} = 0.7068 \text{ Btu/s}$ $= 42.41 \text{ Btu/min} = 2544.5 \text{ Btu/h}$ $= 0.74570 \text{ kW}$ $1 \text{ boiler hp} = 33,475 \text{ Btu}$ $1 \text{ Btu/h} = 1.055056 \text{ kJ/h}$ $1 \text{ ton of refrigeration} = 200 \text{ Btu/min}$
Pressure	$1 \text{ Pa} = 1 \text{ N/m}^2$ $1 \text{ kPa} = 10^3 \text{ Pa} = 10^{-3} \text{ MPa}$ $1 \text{ atm} = 101.325 \text{ kPa} = 1.01325 \text{ bars}$ $= 760 \text{ mm Hg at } 0^\circ\text{C}$ $= 1.03323 \text{ kgf/cm}^2$ $1 \text{ mm Hg} = 0.1333 \text{ kPa}$	$1 \text{ Pa} = 1.4504 \times 10^{-4} \text{ psia}$ $= 0.020886 \text{ lbf/ft}^2$ $1 \text{ psi} = 144 \text{ lbf/ft}^2 = 6.894757 \text{ kPa}$ $1 \text{ atm} = 14.696 \text{ psia} = 29.92 \text{ in Hg at } 30^\circ\text{F}$ $1 \text{ in Hg} = 3.387 \text{ kPa}$
Specific heat	$1 \text{ kJ/kg}\cdot^\circ\text{C} = 1 \text{ kJ/kg}\cdot\text{K} = 1 \text{ J/g}\cdot^\circ\text{C}$	$1 \text{ Btu/lbm}\cdot^\circ\text{F} = 4.1868 \text{ kJ/kg}\cdot^\circ\text{C}$ $1 \text{ Btu/lbmol}\cdot\text{R} = 4.1868 \text{ kJ/kmol}\cdot\text{K}$ $1 \text{ kJ/kg}\cdot^\circ\text{C} = 0.23885 \text{ Btu/lbm}\cdot^\circ\text{F}$ $= 0.23885 \text{ Btu/lbm}\cdot\text{R}$

*Exact conversion factor between metric and English units.

[†]Calorie is originally defined as the amount of heat needed to raise the temperature of 1 g of water by 1°C, but it varies with temperature. The international steam table (IT) calorie (generally preferred by engineers) is exactly 4.1868 J by definition and corresponds to the specific heat of water at 15°C. The thermochemical calorie (generally preferred by physicists) is exactly 4.184 J by definition and corresponds to the specific heat of water at room temperature. The difference between the two is about 0.06 percent, which is negligible. The capitalized Calorie used by nutritionists is actually a kilocalorie (1000 IT calories).

[‡]Mechanical horsepower. The electrical horsepower is taken to be exactly 746 W.



DIMENSION	METRIC	METRIC/ENGLISH
Specific volume	$1 \text{ m}^3/\text{kg} = 1000 \text{ L/kg} = 1000 \text{ cm}^3/\text{g}$	$1 \text{ m}^3/\text{kg} = 16.02 \text{ ft}^3/\text{lbm}$ $1 \text{ ft}^3/\text{lbm} = 0.062428 \text{ m}^3/\text{kg}$
Temperature	$T(\text{K}) = T(^{\circ}\text{C}) + 273.15$ $\Delta T(\text{K}) = \Delta T(^{\circ}\text{C})$	$T(\text{R}) = T(^{\circ}\text{F}) + 459.67 = 1.8T(\text{K})$ $T(^{\circ}\text{F}) = 1.8T(^{\circ}\text{C}) + 32$ $\Delta T(^{\circ}\text{F}) = \Delta T(\text{R}) = 1.8\Delta T(\text{K})$
Thermal conductivity	$1 \text{ W/m}\cdot^{\circ}\text{C} = 1 \text{ W/m}\cdot\text{K}$	$1 \text{ W/m}\cdot^{\circ}\text{C} = 0.57782 \text{ Btu/h}\cdot\text{ft}\cdot^{\circ}\text{F}$
Velocity	$1 \text{ m/s} = 3.60 \text{ km/h}$	$1 \text{ m/s} = 3.2808 \text{ ft/s} = 2.237 \text{ mi/h}$ $1 \text{ mi/h} = 1.46667 \text{ ft/s}$ $1 \text{ mi/h} = 1.6093 \text{ km/h}$
Volume	$1 \text{ m}^3 = 1000 \text{ L} = 10^6 \text{ cm}^3 (\text{cc})$	$1 \text{ m}^3 = 6.1024 \times 10^4 \text{ in}^3 = 35.315 \text{ ft}^3$ $= 264.17 \text{ gal (U.S.)}$ $1 \text{ U.S. gallon} = 231 \text{ in}^3 = 3.7854 \text{ L}$ $1 \text{ fl ounce} = 29.5735 \text{ cm}^3 = 0.0295735 \text{ L}$ $1 \text{ U.S. gallon} = 128 \text{ fl ounces}$
Volume flow rate	$1 \text{ m}^3/\text{s} = 60,000 \text{ L/min} = 10^6 \text{ cm}^3/\text{s}$	$1 \text{ m}^3/\text{s} = 15,850 \text{ gal/min (gpm)} = 35.315 \text{ ft}^3/\text{s}$ $= 2118.9 \text{ ft}^3/\text{min (cfm)}$

Some Physical Constants

Universal gas constant	$R_u = 8.31447 \text{ kJ/kmol}\cdot\text{K}$ $= 8.31447 \text{ kPa}\cdot\text{m}^3/\text{kmol}\cdot\text{K}$ $= 0.0831447 \text{ bar}\cdot\text{m}^3/\text{kmol}\cdot\text{K}$ $= 82.05 \text{ L}\cdot\text{atm/kmol}\cdot\text{K}$ $= 1.9858 \text{ Btu/lbmol}\cdot\text{R}$ $= 1545.37 \text{ ft}\cdot\text{lbf/lbmol}\cdot\text{R}$ $= 10.73 \text{ psia}\cdot\text{ft}^3/\text{lbmol}\cdot\text{R}$
Standard acceleration of gravity	$g = 9.80665 \text{ m/s}^2$ $= 32.174 \text{ ft/s}^2$
Standard atmospheric pressure	$1 \text{ atm} = 101.325 \text{ kPa}$ $= 1.01325 \text{ bar}$ $= 14.696 \text{ psia}$ $= 760 \text{ mm Hg } (0^{\circ}\text{C})$ $= 29.9213 \text{ in Hg } (32^{\circ}\text{F})$ $= 10.3323 \text{ m H}_2\text{O } (4^{\circ}\text{C})$
Stefan–Boltzmann constant	$\alpha = 5.6704 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$ $= 0.1714 \times 10^{-8} \text{ Btu/h}\cdot\text{ft}^2\cdot\text{R}^4$
Boltzmann's constant	$k = 1.380650 \times 10^{-23} \text{ J/K}$
Speed of light in vacuum	$c_o = 2.9979 \times 10^8 \text{ m/s}$ $= 9.836 \times 10^8 \text{ ft/s}$
Speed of sound in dry air at 0°C and 1 atm	$c = 331.36 \text{ m/s}$ $= 1089 \text{ ft/s}$
Heat of fusion of water at 1 atm	$h_{if} = 333.7 \text{ kJ/kg}$ $= 143.5 \text{ Btu/lbm}$
Enthalpy of vaporization of water at 1 atm	$h_{fg} = 2256.5 \text{ kJ/kg}$ $= 970.12 \text{ Btu/lbm}$